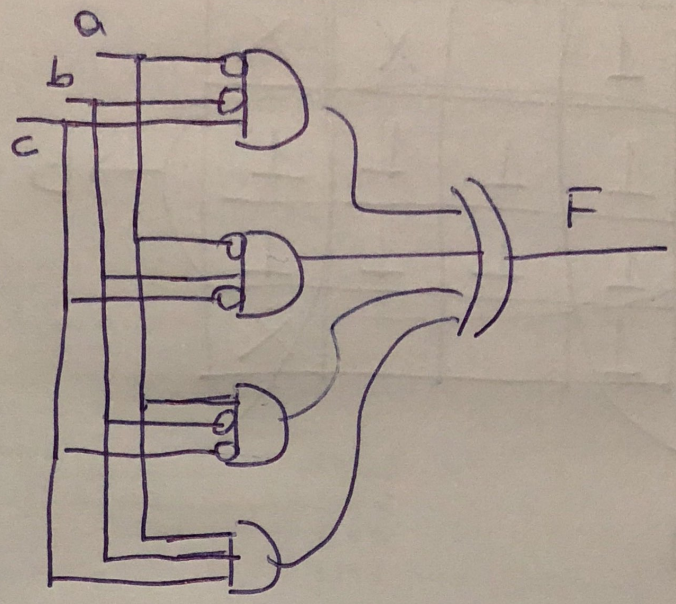
Q1)

a	b	c	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$F = a'b'c + a'bc' + ab'c' + abc$



Q2)

a) $F = ab'$

$G = (a' + ab)'$ — Use distributive law

$= ((a' + a) * (a' + b))' = (a' + b)' = a \cdot b'$

($a' + a = 1$)

b)

a	b	F	G
0	0	0	0
0	1	0	0
1	0	1	1
1	1	0	0

Q3) $wx(x'y + zy' + xy)$

$= wx\,x'y + wx\,zy' + wx\,xy$

($x\,x' = 0$, $x\,x = x$)

$= wx\,zy' + wxy = wx(zy' + y)$

Q5)

AB \ CD	00	01	11	10
00	1		X	X
01	1	1	1	1
11	1	1	1	1
10	1			X

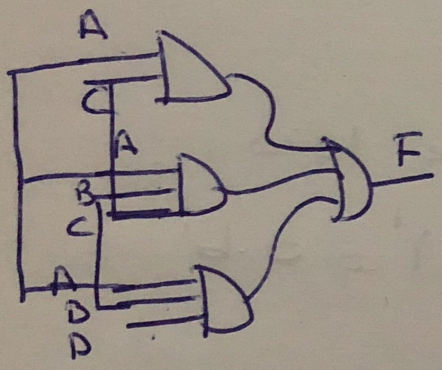
→ D'

→ B

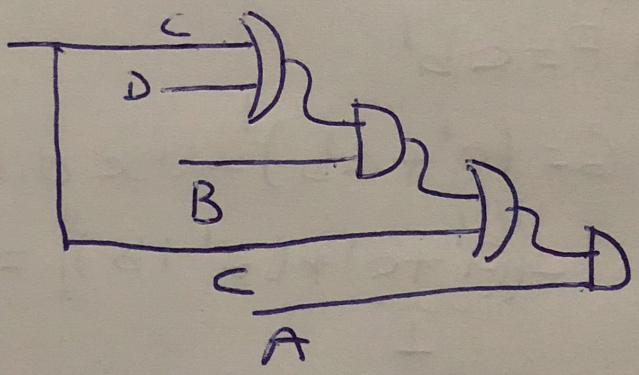
$F = B + D'$

Q6)

$F = AC + ABC + ABD$ $\quad F' = A(C + B(C + D))$



11 * 2 = 22 transistors



8 * 2 = 16 transistors.

Q4) $F(a,b,c) = (a+c)(a'+ab)(b+c)$

$= (a.a' + aab + a'c + abc)(b+c)$

$= abb + abc + a'bc + a'cc + abbc + abcc$

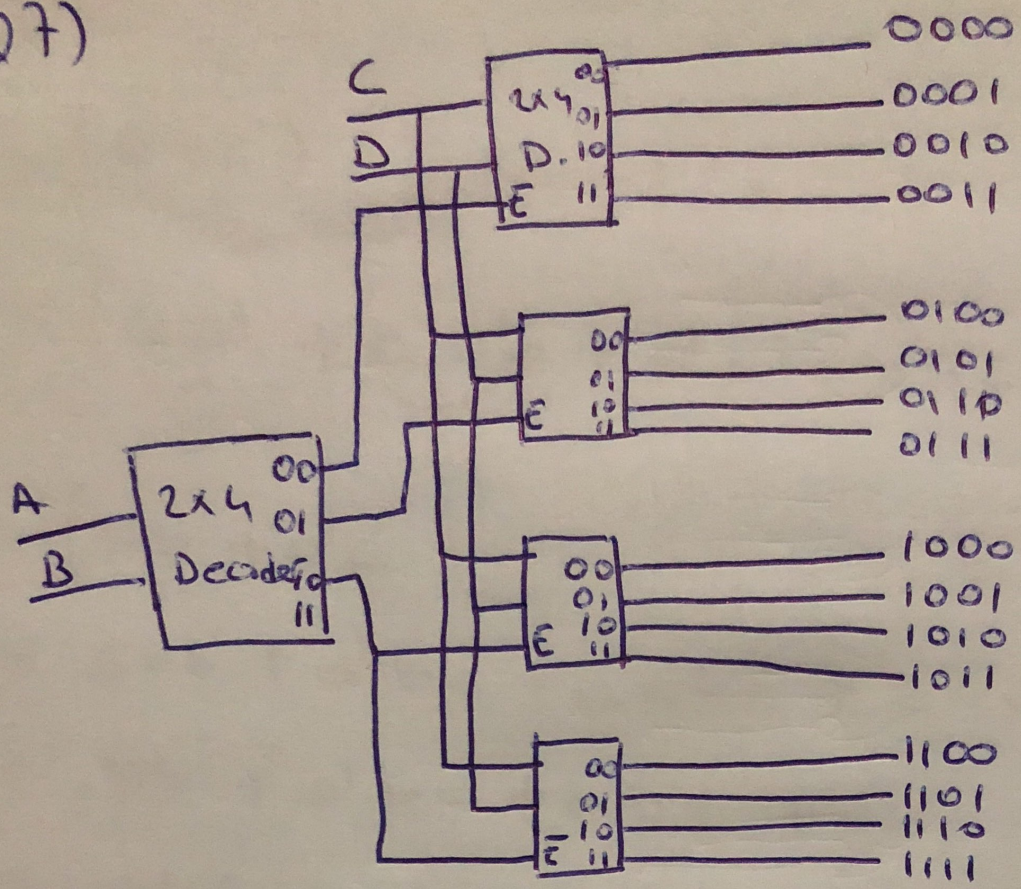
$= ab + abc + a'bc + a'c$

$= ab(c+c') + abc + a'bc + a'c(b+b')$

$= \underbrace{abc}_{7} + \underbrace{abc'}_{6} + \underbrace{a'bc}_{3} + \underbrace{a'b'c}_{1}$

$F = \Sigma m(1,3,6,7)$

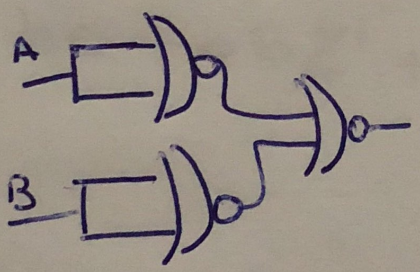
(Q7)



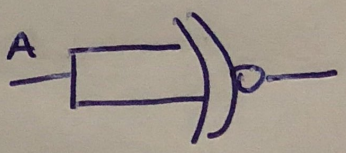
(Q8)

A	B	NOR
0	0	1
0	1	0
1	0	0
1	1	0

AND GATE



NOT GATE



OR GATE

